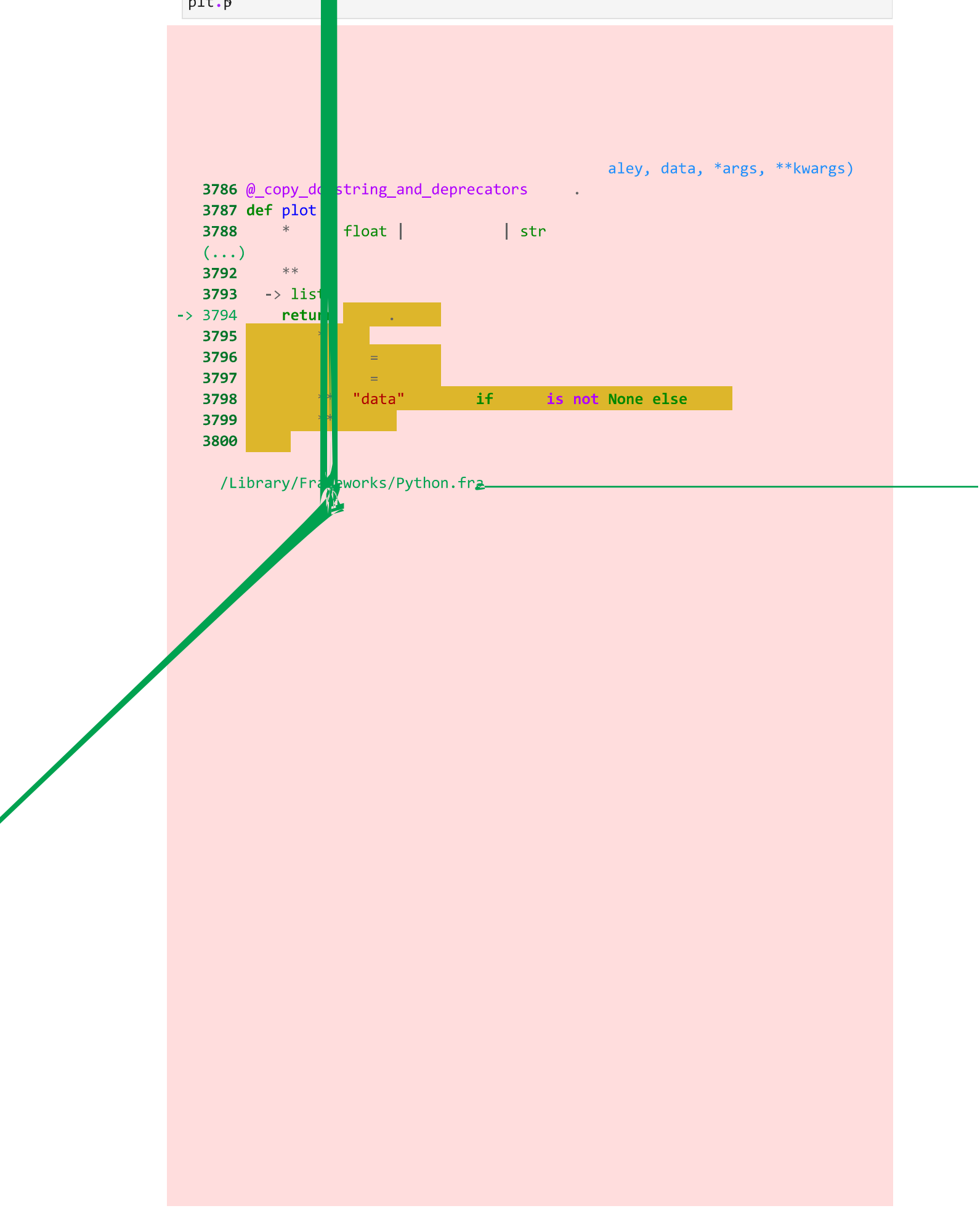
```
#Wake up questions
'''
1. Function to find the odd one in an array. (if others are odd then find the ev

INPUT ARRAY                          OUTPUT
(a)[1,2,3,5,7,9]                        2
(b)[2,4,2,6,8,10,12,14,2,7]             7
Hint: Separate the task of determing the arra as odd or even, based on that find

2. Given a string find the character whose frequency is not like the frequency o

INPUT STRING.                     OUTPUT
(a)aasddffww                        s
(b)asdfghjklpoiuytrewqzxcvbnma      a
'''
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

%matplotlib inline
```

```

```

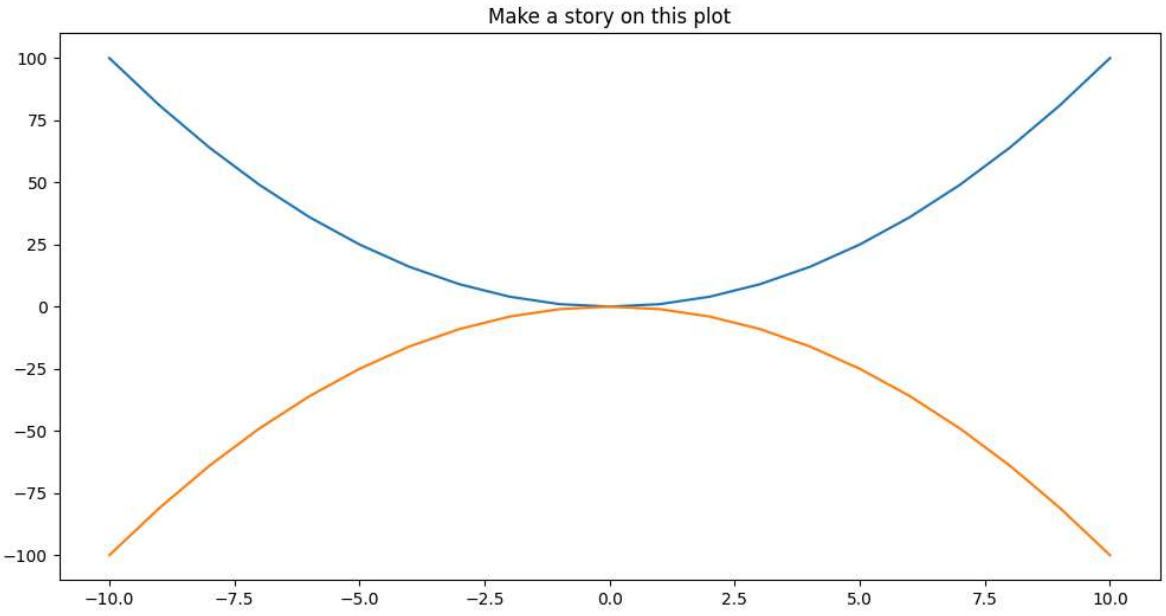
```
x = np.arange(-10, 11)
print(x)
plt.plot(x, x ** 2)
```

```
plt.p
```


aley, data, *args, **kwargs)
3786 @_copy_do string_and_deprecators .
3787 def plot
3788 * float | | str
(...)
3792 **
3793 -> list
-> 3794 retur .
3795 *
3796 =
3797 =
3798 ** "data" if is not None else
3799 **
3800
/Library/Fra eworks/Python.fra

In [7]:
```
plt.figure(figsize=(12, 6))
plt.plot(x, x ** 2)
plt.plot(x, -1 * (x ** 2))

plt.title('Make a story on this plot')
```

Out[7]:



In [32]:
```
df = pd.read_csv('data/btc-market-price.csv')
```

In [33]:
```
df.head()
```

	2017-04-02 00:00:00	1099.169125
0	2017-04-03 00:00:00	1141.813000
1	2017-04-04 00:00:00	1141.600363
2	2017-04-05 00:00:00	1133.079314
3	[illegible]	[illegible]

```
df = pd.read_csv('data/btc-market-price.csv', header=None)
df.head()
```

		1099.169125
1	2017-04-03 00:00:00	1141.813000
2	2017-04-04 00:00:00	1141.600363
3	2017-04-05 00:00:00	1133.079314
4	2017-04-06 00:00:00	1196.307937

In [17]:
```
df.columns = ['Timestamp', 'Price']
```

In [19]:
```
df.shape
```

Out[19]: (365, 2)

In [20]:
```
df.tail()
```

Out[20]:

	Timestamp	Price
360	2018-03-28 00:00:00	7960.380000
361	2018-03-29 00:00:00	7172.280000
362	2018-03-30 00:00:00	6882.531667
363	[illegible]	6935.480000

In [23]:
```
df.head(30)
```

	Timestamp	Price
0	2017-04-02 00:00:00	1099.169125
1	2017-04-03 00:00:00	1141.813000
2	2017-04-04 00:00:00	1141.600363
3	2017-04-05 00:00:00	1133.079314
4	2017-04-06 00:00:00	[illegible]
5	[illegible]	
8	2017-04-10 00:00:00	1207.744875
9	2017-04-11 00:00:00	1226.617037
10	2017-04-12 00:00:00	1218.922050
11	2017-04-13 00:00:00	1180.023713
12	2017-04-14 00:00:00	1185.260057
13	2017-04-15 00:00:00	1184.880671
14	2017-04-16 00:00:00	1186.927412
15	2017-04-17 00:00:00	1205.634875
16	2017-04-18 00:00:00	[illegible]
19	2017-04-21 00:00:00	1258.361412
20	2017-04-22 00:00:00	1261.311225
21	2017-04-23 00:00:00	1257.988112
22	2017-04-24 00:00:00	1262.902775
23	2017-04-25 00:00:00	1279.414687
24	2017-04-26 00:00:00	1309.109875
25	2017-04-27 00:00:00	1345.353912
26	2017-04-28 00:00:00	1331.294429
27	2017-04-29 00:00:00	[illegible]
	[illegible]	

In [24]:
```
df.dtypes
```

```
df.columns = ['Timestamp', 'Price']
df['Timestamp'] = pd.to_datetime(df['Timestamp'])
df.set_index('Timestamp', inplace=True)
df.head()
```

Out[34]:

	Pr
[redacted]	1099.169125
2017-04-03	1141.813000
2017-04-04	1141.600363
2017-04-05	1133.079314
2017-04-06	1196.307937

In [35]:

```
df = pd.read_csv(
    'data/btc-market-price.csv',
    header=None,
    names=['Timestamp', 'Price'],
    index_col=0,
    parse_dates=True
)
df.head()
```

Out[35]:

	Price
Timestamp	
2017-04-02	1099.169125
2017-04-03	1141.813000
2017-04-04	1141.600363
2017-04-05	1133.079314
2017-04-06	1196.307937

In [36]:

```
df.loc['2017-09-29']
```

Out[36]:

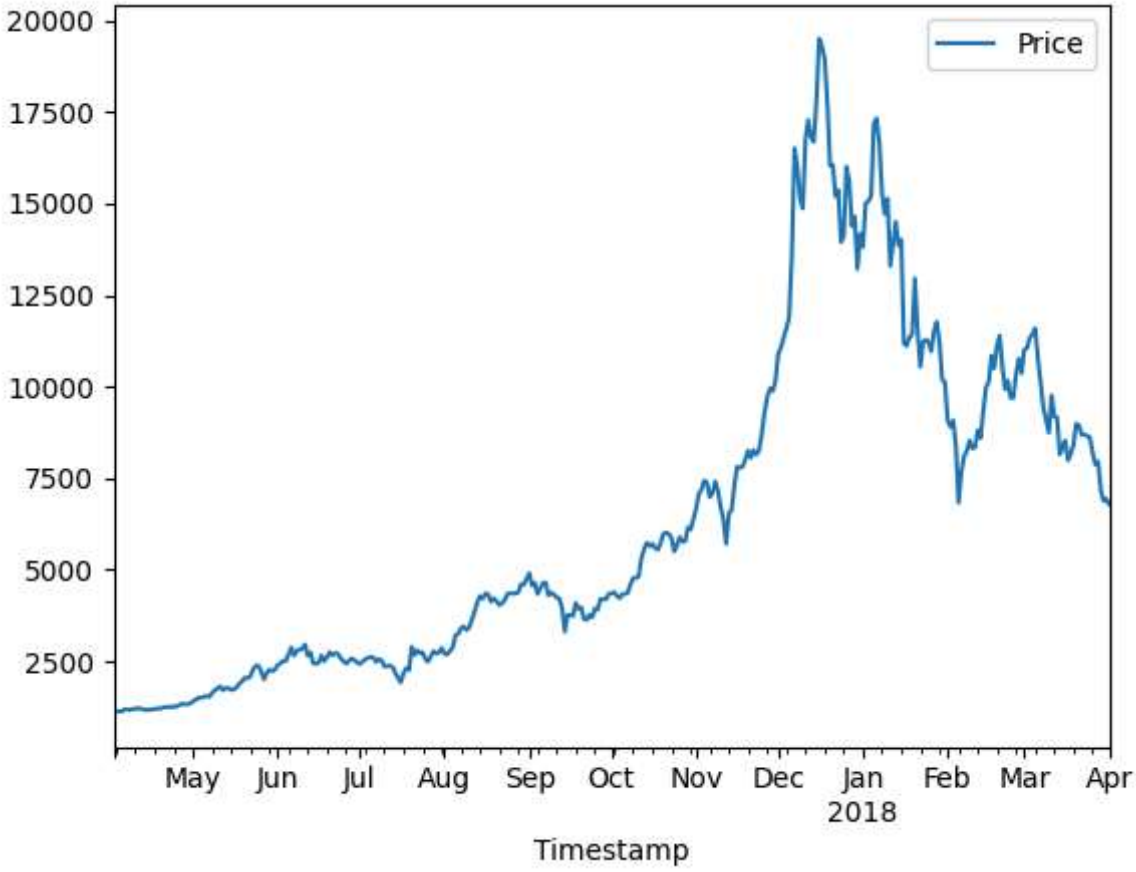
```
Price    4193.574667
Name: 2017-09-29 00:00:00, dtype: float64
```

In [39]:

```
df.plot()
```

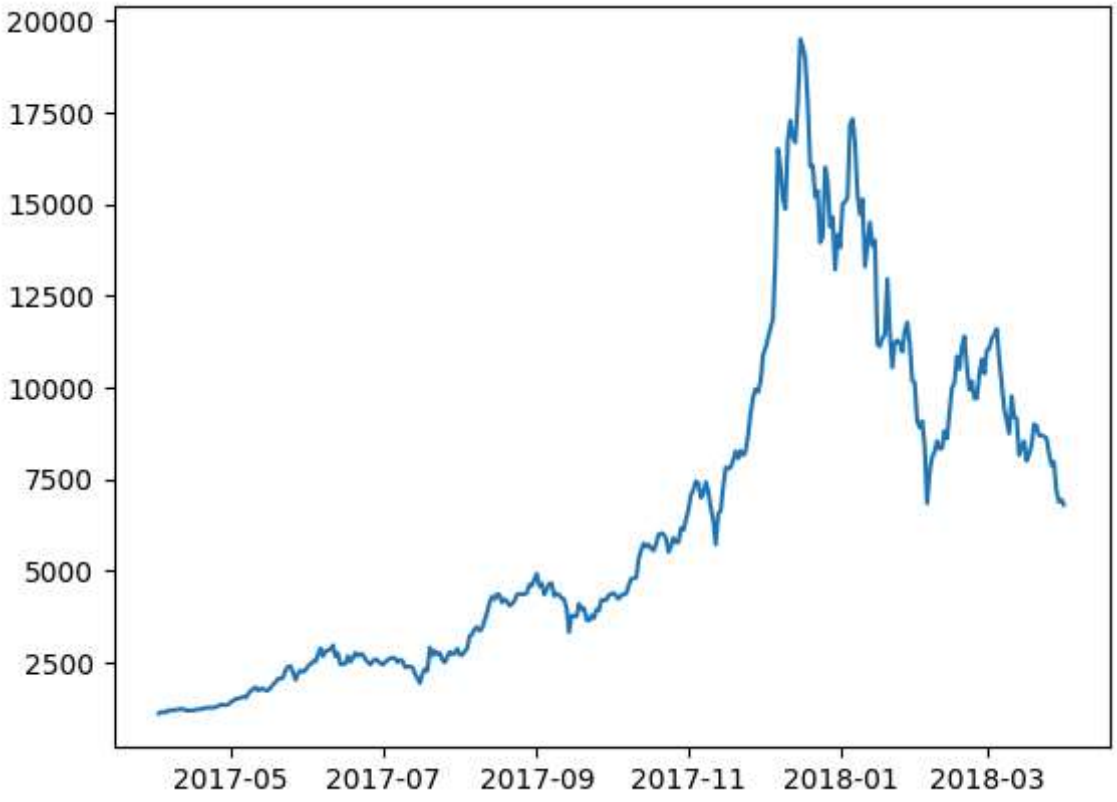
Out[39]:

```
<Axes: xlabel='Timestamp'>
```



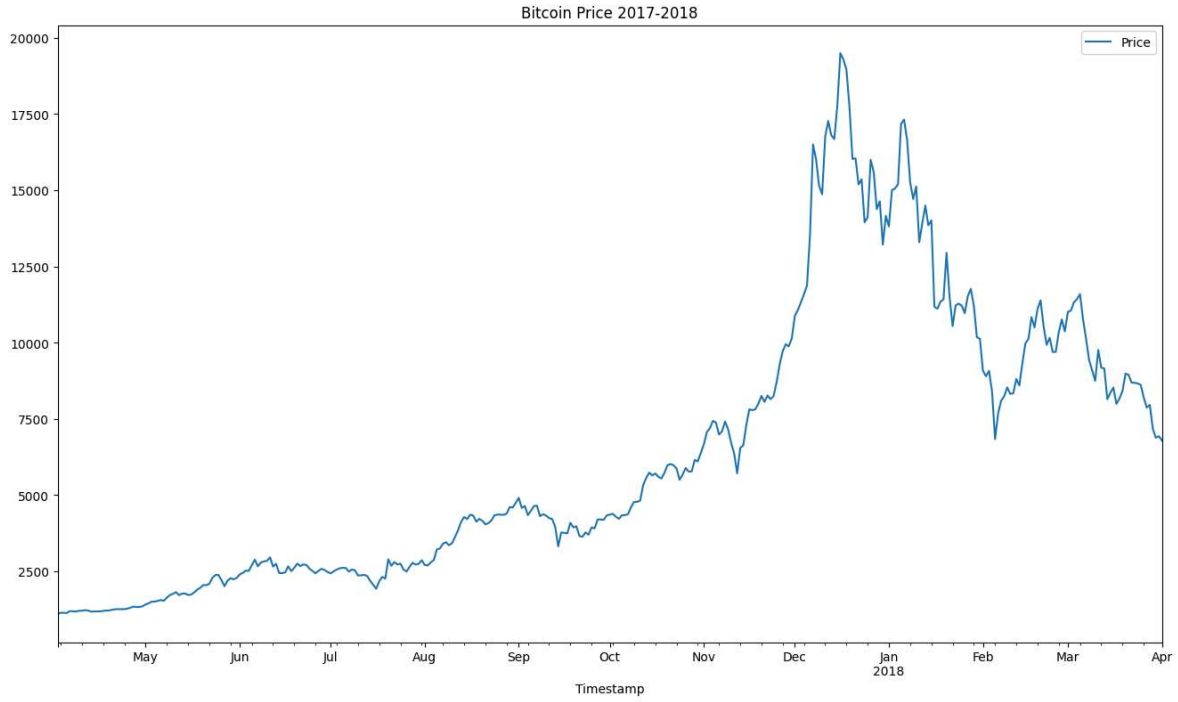
In [40]:
```
plt.plot(df.index, df['Price'])
```

Out[40]: [<matplotlib.lines.Line2D at 0x129b5a950>]



In [41]:
```
df.plot(figsize=(16, 9), title='Bitcoin Price 2017-2018')
```

Out[41]: <Axes: title={'center': 'Bitcoin Price 2017-2018'}, xlabel='Timestamp'>



In [ ]:

In [ ]:

In [42]:
```
# flash back to python - find the number of days when the stock price was greate
c = 0
for p in df["Price"]:
    #code here
    if p > 5000:
        c += 1
print(c)
```

```
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```

In [49]:
```
df["5000 pivot"] = df["Price"]>5000
df["Price 2"] = df["Price"]
```

In [46]:
```
df["5000 pivot"].value_counts()
```

Out[46]:
```
5000 pivot
False    193
True     172
Name: count, dtype: int64
```

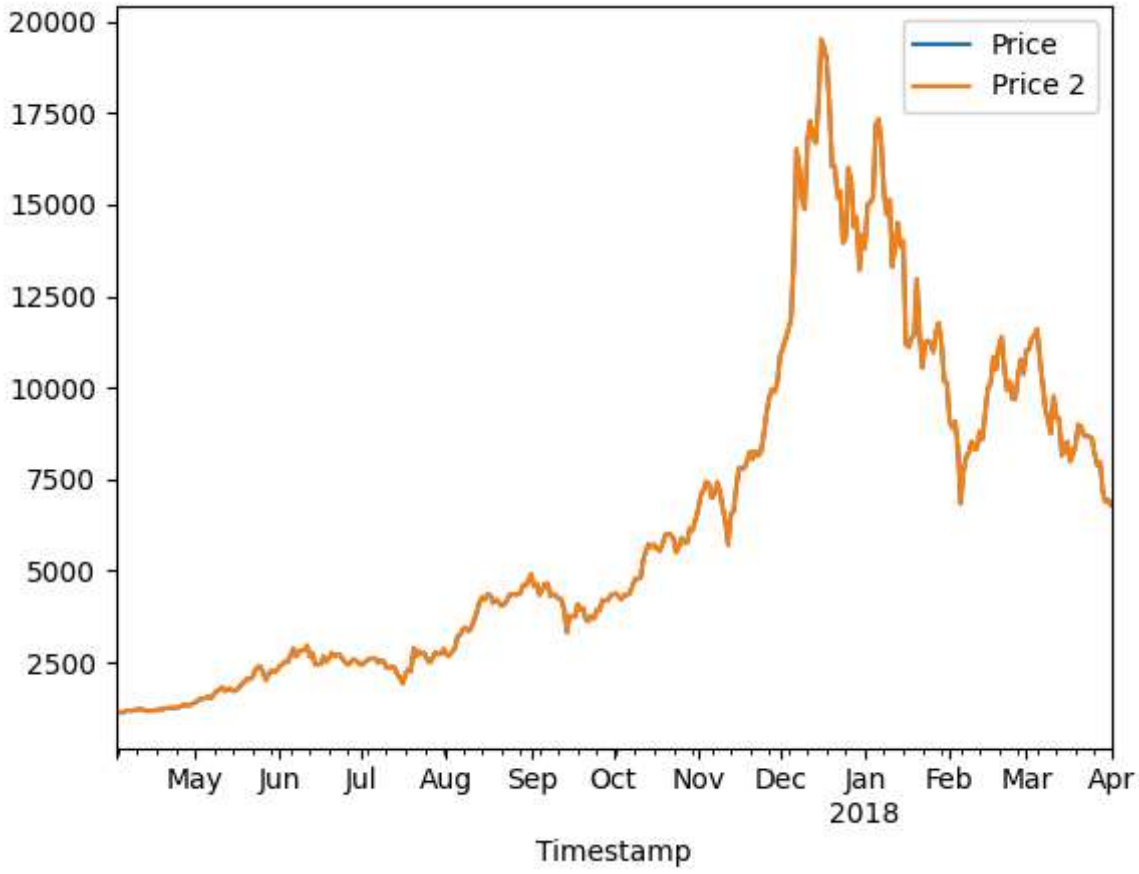
In [50]:
```
df.head()
```

Out[50]:

Timestamp	Price	5000 pivot	Price 2
2017-04-02	1099.169125	False	1099.169125
2017-04-03	1141.813000	False	1141.813000
2017-04-04	1141.600363	False	1141.600363
2017-04-05	1133.079314	False	1133.079314
2017-04-06	1196.307937	False	1196.307937

In [51]:
```
df.plot()
```

Out[51]:
```
<Axes: xlabel='Timestamp'>
```



In [53]:
```
c = (df["Price"] > 5000).sum()
print(c)
```

```
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```

In [ ]:
```
# Homework
1. Find out what interface the df.plot() uses at the back end.
```